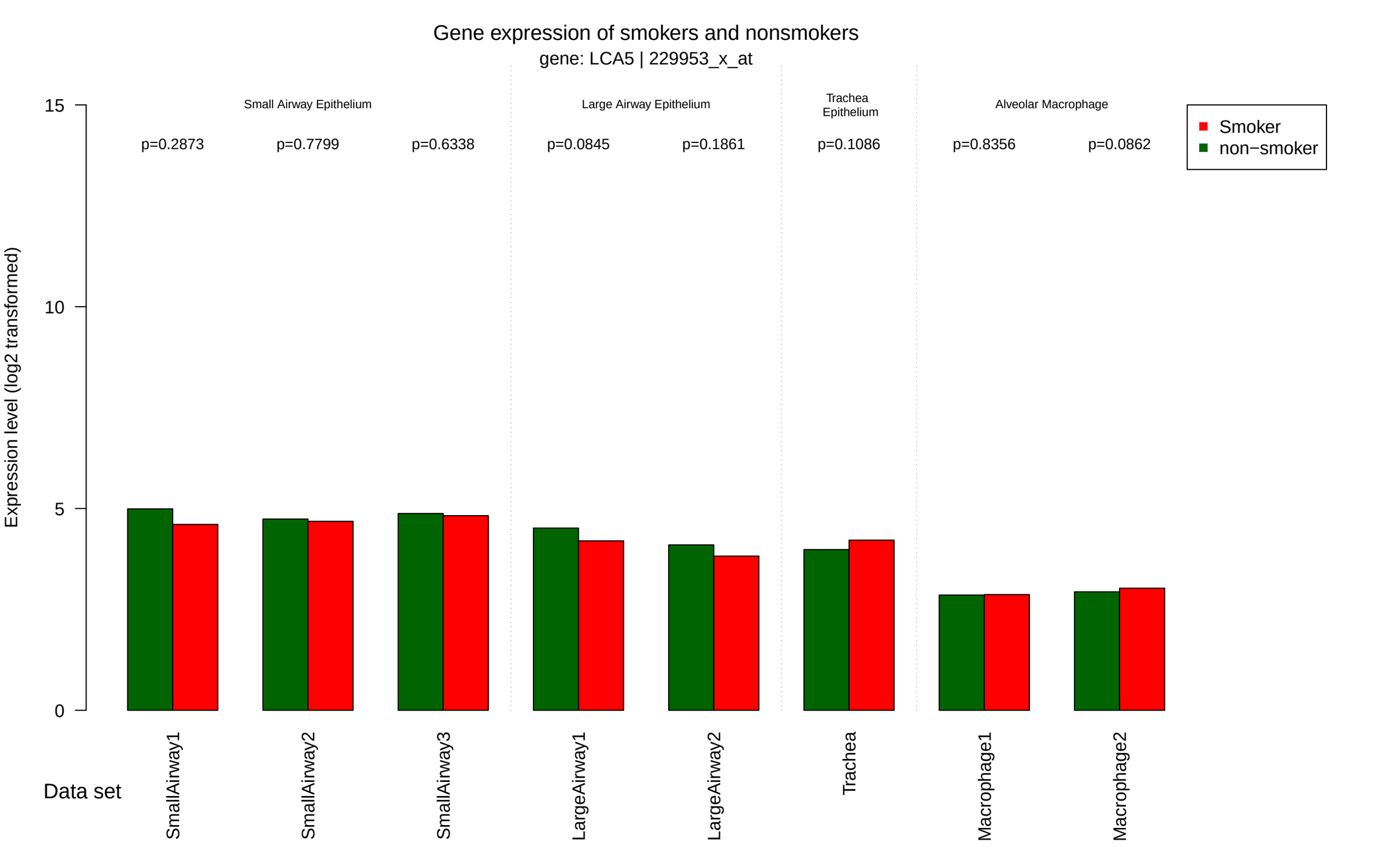
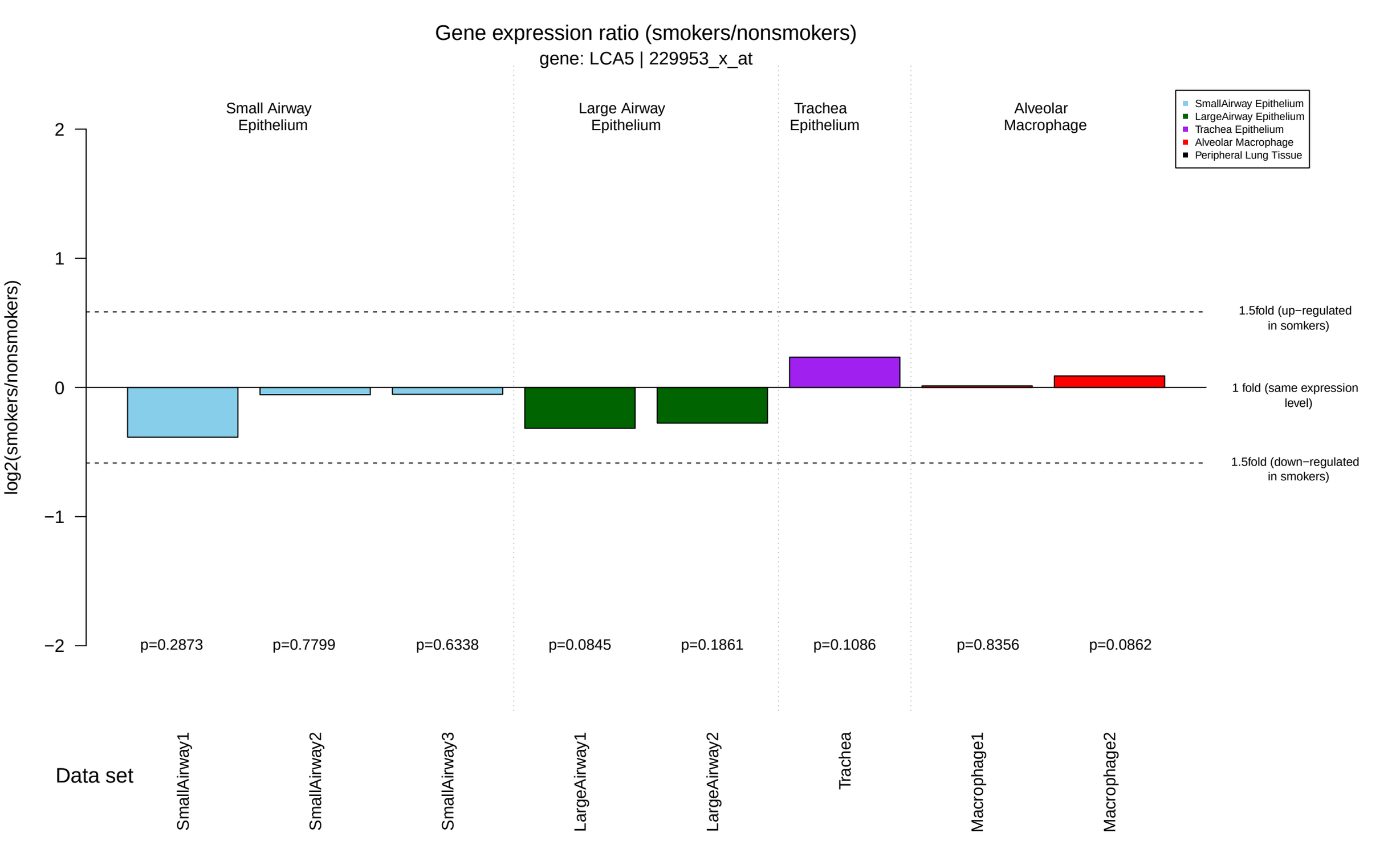
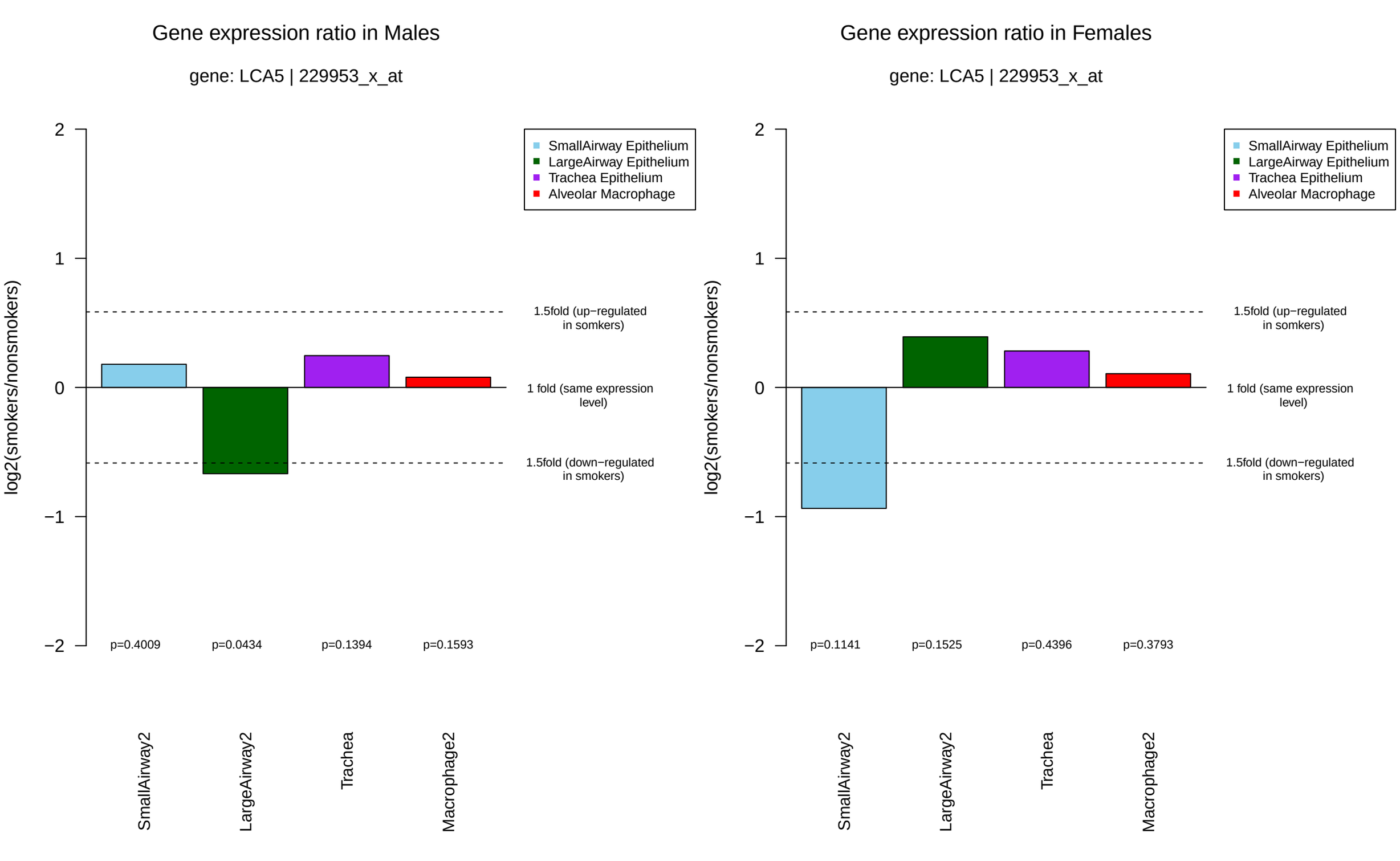
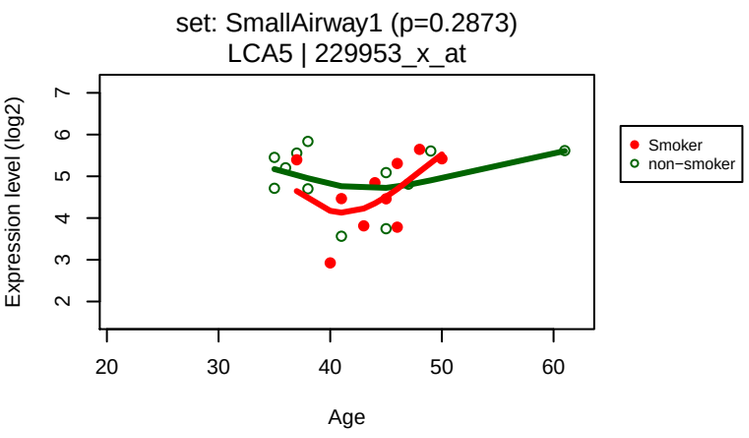
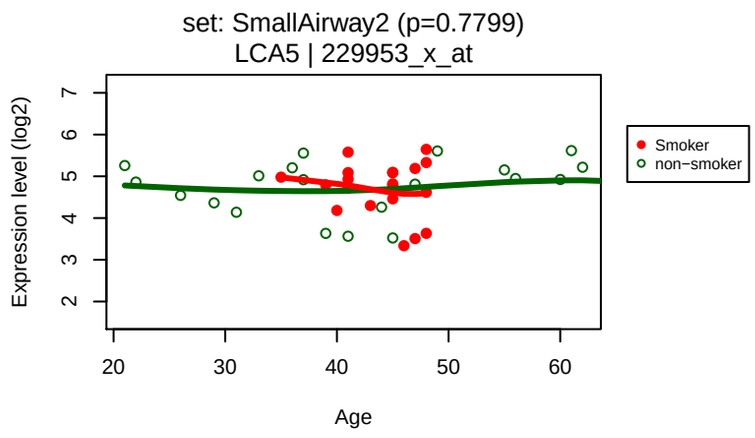
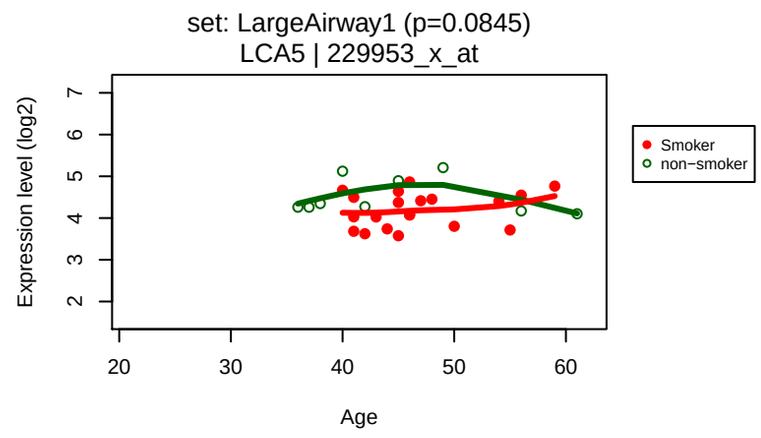

Gene expression of smokers and nonsmokers
gene: LCA5 | 229953_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2873
p=0.7799
p=0.6338
p=0.0845
p=0.1861
p=0.1086
p=0.8356
p=0.0862
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

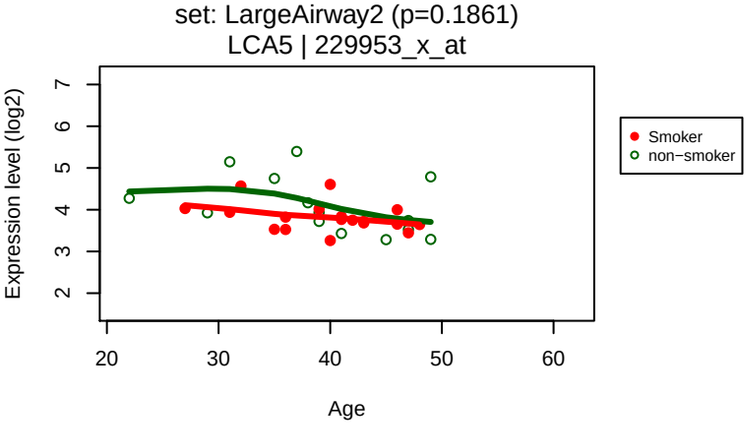

Gene expression ratio (smokers/nonsmokers)
gene: LCA5 | 229953_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2873
p=0.7799
p=0.6338
p=0.0845
p=0.1861
p=0.1086
p=0.8356
p=0.0862
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

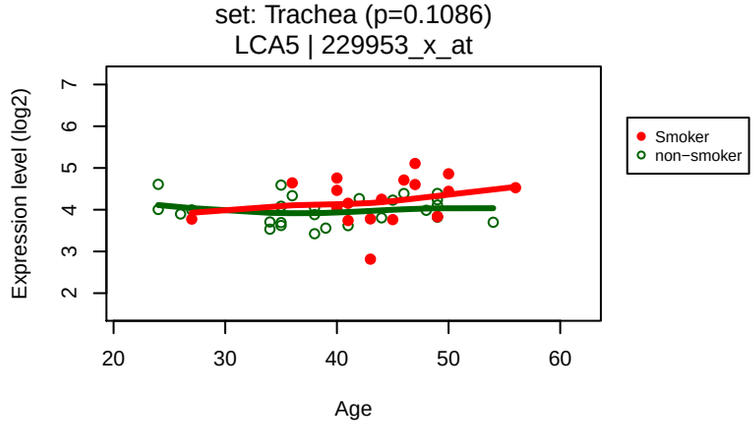

Gene expression ratio in Males
gene: LCA5 | 229953_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4009
p=0.0434
p=0.1394
p=0.1593
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LCA5 | 229953_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1141
p=0.1525
p=0.4396
p=0.3793
SmallAirway2
LargeAirway2
Trachea
Macrophage2

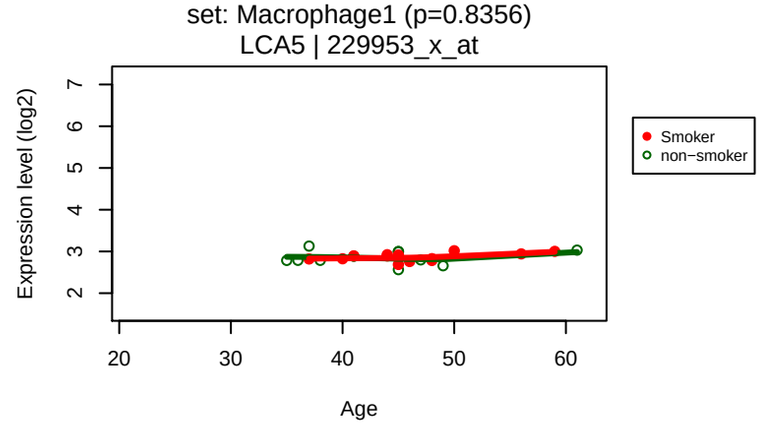

set: SmallAirway1 (p=0.2873)
LCA5 | 229953_x_at
Expression level (log2)
Age
Smoker
non−smoker

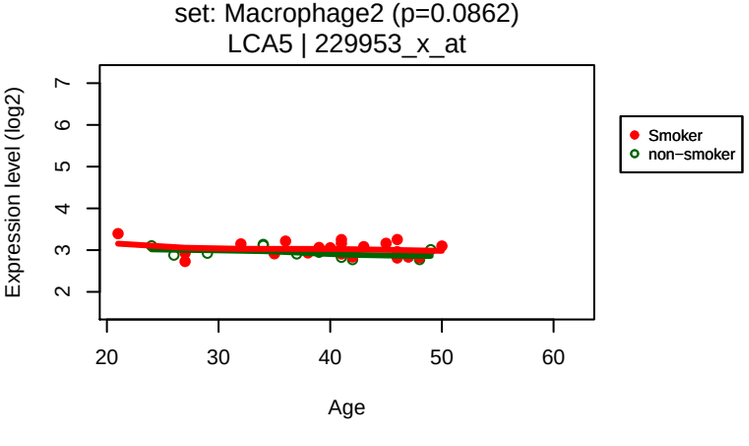

set: SmallAirway2 (p=0.7799)
LCA5 | 229953_x_at
Expression level (log2)
Age
Smoker
non−smoker

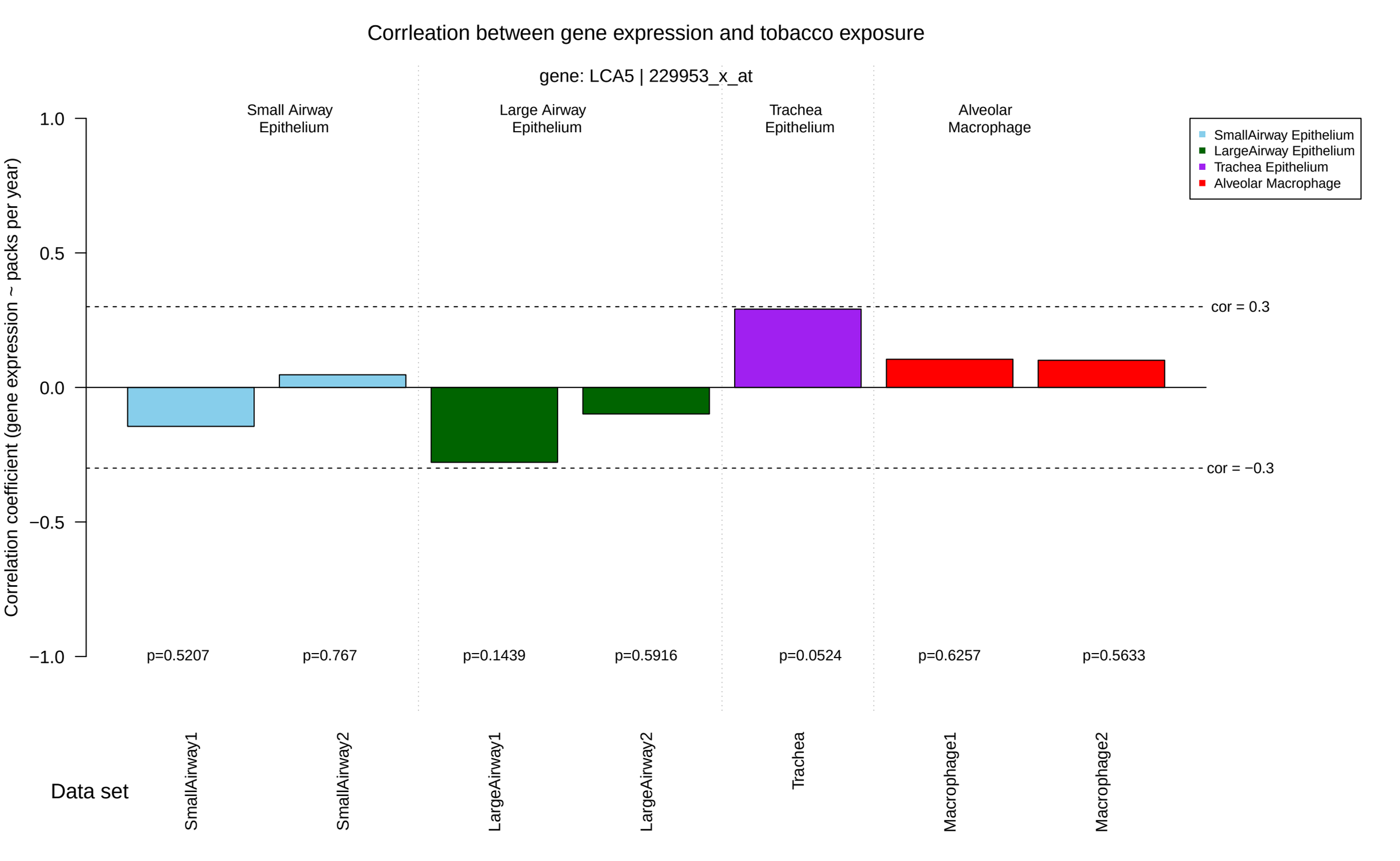

set: LargeAirway1 (p=0.0845)
LCA5 | 229953_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1861)
LCA5 | 229953_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1086)
LCA5 | 229953_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8356)
LCA5 | 229953_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0862)
LCA5 | 229953_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LCA5 | 229953_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5207
p=0.767
p=0.1439
p=0.5916
p=0.0524
p=0.6257
p=0.5633
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2